

Week 1 Final Project Report

Adult Census Income Classification

1. Problem Definition

The objective of this project was to develop a machine learning system capable of predicting whether an individual's annual income exceeds \$50K. This type of binary classification problem has practical relevance in areas such as targeted marketing, customer segmentation, and resource allocation, where organizations may want to identify individuals with characteristics associated with higher economic value.

For this project, the positive class was defined as individuals earning more than \$50K per year. The primary evaluation metric selected was precision. The business motivation for this choice was to minimize false-positive predictions. In a targeted outreach scenario, incorrectly identifying a person as a high-value target could result in wasted marketing resources. Therefore, the project prioritized the reliability of positive predictions, while accepting a trade-off in recall.

The complete Week 1 workflow covered data preparation, exploratory analysis, baseline development, supervised learning, feature engineering, cross-validation, model tuning, probability calibration, threshold selection, final validation, model interpretation, and production-ready inference.

2. Data Preparation and Validation Strategy

The UCI Adult Census Income dataset was used for this project and accessed through OpenML using Scikit-learn. The dataset contains demographic, educational, employment, and financial information, including age, workclass, education, marital status, occupation, capital gain, capital loss, hours worked per week, and native country.

Missing values represented by ? were converted to NaN. The target variable was converted into a binary format where 1 represented income greater than \$50K and 0 represented income less than or equal to \$50K.

A reproducible 80/20 stratified train/test split was created using `random_state=42`. Stratification ensured that the class distribution remained similar across the training and test datasets. The hold-out test set was reserved for final evaluation and was not used during model training or hyperparameter optimization.

For preprocessing, numeric features were processed using median imputation and standard scaling. Median imputation was selected because it is more robust to extreme values than mean imputation. Categorical features were processed using missing-value handling and one-hot encoding with unknown categories ignored. All preprocessing was included inside the machine learning pipeline to reduce data leakage and ensure that the same transformations could be applied consistently during inference.

Feature engineering was also explored during the project. New features were created using information available within each individual observation, including age groups, working-hours groups, capital-gain indicators, logarithmic transformations of financial variables, higher-education indicators, and feature interactions. No target encoding was used, preventing the engineered features from leaking target information.

3. Model Development and Baseline Comparison

The project began with simple baseline models to establish a minimum performance reference.

The majority-class baseline always predicted the most frequent class. Because this strategy did not predict the positive income class, it achieved zero precision for the positive class.

A rule-based baseline was also tested and achieved a precision of approximately 0.4844. Although this was better than the majority-class predictor, it demonstrated that a simple manually designed rule was insufficient for achieving highly reliable positive predictions.

Several supervised learning models were then evaluated, including Logistic Regression, Random Forest, and HistGradientBoosting. Cross-validation results showed precision values of approximately 0.7455 for Logistic Regression, 0.7230 for Random Forest, and 0.7781 for HistGradientBoosting.

HistGradientBoosting achieved the strongest precision among the main supervised candidates and was selected for further development. Logistic Regression remained useful as an interpretable reference model, while tree-based models provided the ability to capture nonlinear relationships and interactions between features.

4. Model Tuning and Diagnostics

The selected candidate models were further developed using reproducible machine learning pipelines and controlled hyperparameter optimization. Stratified cross-validation was used during the search process, and precision remained the primary optimization metric.

Model diagnostics were performed to investigate potential overfitting and underfitting. Learning curves were used to compare training and validation performance across different training sizes. Regularization and model complexity were also considered during the tuning process.

Probability calibration was performed because the final model was intended to generate probability estimates that could be used for business threshold selection. The Brier score was used as an additional measure of probability quality.

Instead of relying on the default classification threshold of 0.50, multiple thresholds were evaluated. A threshold of 0.80 was selected because it produced the strongest alignment with the precision-focused business objective. Increasing the threshold made the model more selective before predicting the positive class, substantially reducing false positives while also reducing recall.

5. Final Results

The final tuned and calibrated HistGradientBoosting pipeline was evaluated on the untouched hold-out test set.

The final results were:

Metric	Score
Accuracy	0.8372
Precision	0.9675
Recall	0.3311
F1 Score	0.4933
ROC-AUC	0.9226
PR-AUC	0.8211
Brier Score	0.0904

The final model achieved a precision of 96.75%, meaning that positive predictions were highly reliable. The ROC-AUC of 0.9226 and PR-AUC of 0.8211 also indicate strong discrimination and ranking ability.

The primary trade-off was recall. At the selected threshold, the model identified approximately 33.11% of the actual high-income individuals. This behavior was expected because the project deliberately prioritized precision.

The final confusion matrix contained:

- 7,405 True Negatives
- 26 False Positives
- 1,564 False Negatives
- 774 True Positives

The extremely small number of false positives demonstrates the effectiveness of the high threshold for the selected business objective. However, the large number of false negatives confirms that many actual high-income individuals were not identified.

6. Error Analysis and Model Behavior

Detailed error analysis showed that false negatives were the dominant error type. This was directly related to the precision-focused threshold of 0.80.

False positives generally displayed characteristics associated with higher income. On average, they were older, had higher education levels, higher capital gains, and worked more hours per week than the false-negative group. This suggests that the model sometimes

encounters individuals whose observable characteristics strongly resemble the positive class even though their actual income is below \$50K.

False negatives often had weaker financial signals and lower education numbers than correctly identified positive cases. However, many false negatives still had Bachelor's degrees, professional occupations, or substantial working hours. This demonstrates that income cannot be fully explained by the features available in the dataset.

Subgroup analysis was also performed across sex, age groups, and education categories. Precision remained high across the major sex groups, although recall varied. Younger individuals showed particularly low recall, while the senior age group achieved stronger recall and F1 performance. Performance also varied across education categories, demonstrating that education interacts with occupation, age, financial characteristics, and other variables.

These results highlight the importance of monitoring model behavior across different groups, especially if the system were used in a real-world environment.

7. Model Interpretation

Because the final model was a calibrated HistGradientBoosting pipeline, permutation feature importance was used for interpretation.

The most influential features were:

1. Marital status
2. Capital gain
3. Education number
4. Capital loss
5. Workclass
6. Occupation

Marital status and capital gain produced the largest decreases in precision when their values were randomly shuffled, indicating that the final model relied heavily on these variables.

Education number also provided substantial predictive information, while employment-related variables such as workclass and occupation contributed additional signals.

Feature importance represents predictive influence rather than causation. In particular, demographic and socioeconomic variables may reflect historical patterns and inequalities present in the dataset. Therefore, these features should be interpreted carefully in any real-world application.

8. Production Readiness

The final trained machine learning pipeline was saved as a Joblib artifact named `final_model.joblib`.

The saved pipeline enables raw input data to be passed directly to the model. The pipeline automatically applies the transformations required before generating predictions, reducing the risk of manually applying inconsistent preprocessing during deployment.

The production inference workflow:

1. Loads the saved model artifact.
2. Accepts new raw observations.
3. Generates positive-class probabilities.
4. Applies the selected classification threshold of 0.80.
5. Returns the final binary prediction.

The inference workflow was tested using ten unseen examples from the hold-out dataset. The pipeline successfully generated probabilities and predictions without manually repeating preprocessing.

This structure provides a foundation for future deployment through a REST API, web application, or automated prediction workflow.

9. Limitations and Future Improvements

The project has several important limitations.

First, the Adult Census Income dataset is historical and may not represent modern economic conditions. Second, income is influenced by many factors that are not included in the dataset, such as detailed work experience, industry conditions, geographic location, and current labor-market dynamics.

Third, the selected threshold produces excellent precision but relatively low recall. If the business objective changes and identifying a larger proportion of high-income individuals becomes more important, the threshold should be lowered or optimized using a different cost function.

Fourth, demographic attributes may reflect historical social and economic inequalities. A more complete fairness analysis would be required before using the model in a high-impact decision-making environment.

Future improvements could include testing advanced gradient boosting frameworks such as XGBoost, LightGBM, or CatBoost, performing Bayesian hyperparameter optimization, adding more domain-informed features, using SHAP for deeper model explainability, and implementing production monitoring for data drift.

Final Conclusion

This Week 1 project successfully developed a complete end-to-end machine learning solution for Adult Census Income classification.

The workflow progressed from exploratory data analysis and simple baseline models to supervised learning, feature engineering, cross-validation, hyperparameter tuning, calibration, threshold optimization, final validation, error analysis, model interpretation, and production-ready inference.

The final calibrated HistGradientBoosting model achieved a precision of 96.75% on the untouched hold-out test set using a classification threshold of 0.80. Although this resulted in lower recall, the outcome aligns with the selected business objective of minimizing false-positive predictions.

The final project demonstrates a reproducible machine learning workflow and provides a complete foundation for further experimentation, deployment, monitoring, and future model improvement.